

◆ WEEK 2 — SESSION 5

GITHUB LAB GUIDE

STORAGE LAYER — AMAZON S3

[Amazon S3](#) • [Object Storage](#) • [Storage Classes](#) • [Storage Security](#)

◆ LAB OBJECTIVE

In this lab, students will:

- ✓ Create an Amazon S3 bucket
 - ✓ Upload and manage files
 - ✓ Understand object storage
 - ✓ Explore S3 storage classes
 - ✓ Configure bucket permissions
 - ✓ Understand public vs private access
 - ✓ Analyze storage security risks
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◆ REAL-WORLD ENGINEERING GOAL

This lab simulates how cloud engineers manage storage infrastructure in enterprise environments.

Students must think about:

- ◆ scalability
 - ◆ security
 - ◆ governance
 - ◆ access control
 - ◆ and operational risk
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◆ REQUIRED TOOLS

Before starting, ensure you have:

- ✓ AWS Account
- ✓ Internet Browser
- ✓ GitHub Account
- ✓ VS Code (Recommended)
- ✓ Stable internet connection

◆ GITHUB REPOSITORY STRUCTURE

Recommended Repository Name

aws-s3-storage-lab

◆ STEP 1 — CREATE GITHUB REPOSITORY

Instructions

1. Login to GitHub
2. Click:
 - ◆ New Repository
3. Repository Name:

aws-s3-storage-lab

4. Add:

✓ README.md

5. Click:
 - ◆ Create Repository
-

◆ STEP 2 — CREATE PROJECT FOLDER

Recommended Folder Structure

aws-s3-storage-lab/

```
|  
├── screenshots/  
├── notes/  
├── configurations/  
└── README.md
```

◆ STEP 3 — LOGIN TO AWS

Instructions

1. Open AWS Console
2. Login using IAM User
3. Navigate to:

Amazon S3

◆ STEP 4 — CREATE S3 BUCKET

Instructions

1. Click:

Create Bucket

2. Enter bucket name

Example:

mz-storage-lab-yourname

⚠ Bucket names must be globally unique.

◆ IMPORTANT UNDERSTANDING

An S3 bucket is a container that stores objects.

Objects can include:

- ◆ images
 - ◆ videos
 - ◆ documents
 - ◆ backups
 - ◆ application files
-

◆ STEP 5 — CONFIGURE BUCKET SETTINGS

Recommended Configuration

Bucket Type

✓ General Purpose

Block Public Access

✓ Keep enabled initially

Bucket Versioning

Optional:
Enable for learning purposes

◆ SECURITY THINKING

Why does AWS block public access by default?

Because accidental public exposure can leak sensitive data.

◆ STEP 6 — UPLOAD FILES

Instructions

1. Open bucket
2. Click:

Upload

3. Upload:

- ✓ image
 - ✓ PDF
 - ✓ text file
-

◆ OBJECT STORAGE UNDERSTANDING

Each uploaded file becomes an:

Object

Each object contains:

- ◆ file data
 - ◆ metadata
 - ◆ unique identifier
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◆ STEP 7 — CREATE STORAGE CLASS TEST

Instructions

Upload multiple files using different storage classes.

Test:

- ✓ Standard
 - ✓ Standard-IA
 - ✓ Intelligent-Tiering
-

◆ REQUIRED OBSERVATION

Students must compare:

- ◆ pricing differences
 - ◆ access patterns
 - ◆ intended business use cases
-

◆ STORAGE CLASS THINKING

Ask yourself:

Why would businesses archive old data instead of storing everything in expensive high-speed storage?

◆ STEP 8 — TEST PUBLIC ACCESS

⚠ IMPORTANT:

This section is for educational purposes only.

Instructions

1. Select uploaded file
2. Open:

Permissions

3. Temporarily allow public access

4. Generate object URL
5. Test browser access

◆ REQUIRED THINKING

Students must answer:

- ◆ What risks exist if sensitive files are public?
- ◆ Why is unrestricted access dangerous?
- ◆ What could attackers discover?

◆ STEP 9 — RESTRICT ACCESS AGAIN

Instructions

Immediately:

- ✓ remove public access
- ✓ restore secure settings

◆ SECURITY PRINCIPLE

Good cloud engineers test exposure safely — then secure systems immediately.

◆ STEP 10 — DOCUMENT FINDINGS

Create:

`notes/storage-security-analysis.md`

◆ REQUIRED ANALYSIS QUESTIONS

Students must answer:

1. What is Amazon S3?

2. What is object storage?

3. What risks exist with public buckets?

4. Why do storage classes exist?

5. What storage class would fit backups?

6. Why must cloud storage be governed carefully?

◆ STEP 11 — TAKE SCREENSHOTS

Save screenshots inside:

```
screenshots/
```

Required screenshots:

- ✓ Bucket creation
 - ✓ Uploaded files
 - ✓ Storage classes
 - ✓ Permissions configuration
 - ✓ Public access settings
-

◆ STEP 12 — UPDATE README.md

Recommended README Structure

```
# AWS S3 Storage Lab

## Objective
This lab demonstrates:
- Amazon S3
- Object storage
- Storage classes
- Cloud storage security

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## Tasks Completed
- Created S3 bucket
- Uploaded objects
- Tested storage classes
- Configured permissions
- Analyzed storage risks

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```

```
## Key Learnings
- S3 uses object storage
- Public storage creates risk
- Storage classes optimize cost
- Governance is critical in cloud storage
```

```
## Security Reflection
Misconfigured cloud storage can expose sensitive enterprise data publicly.
```

◆ STEP 13 — PUSH TO GITHUB

Commands

```
git init
git add .
git commit -m "Completed AWS S3 Storage Lab"
git branch -M main
git remote add origin YOUR_REPOSITORY_URL
git push -u origin main
```

◆ FINAL SUBMISSION REQUIREMENTS

Students must submit:

- ✓ GitHub repository link
 - ✓ Screenshots
 - ✓ Storage analysis notes
 - ✓ README documentation
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◆ ENTERPRISE THINKING SECTION

Important Reality

Some of the largest cloud breaches globally happened because:

- ◆ storage was public,
- ◆ permissions were weak,
- ◆ or governance was poorly managed.

Cloud storage security is a serious engineering responsibility.

◆ BONUS CHALLENGE (ADVANCED)

Students may optionally:

- ✓ enable bucket versioning
 - ✓ create lifecycle rules
 - ✓ explore Glacier storage
 - ✓ test encryption settings
-

◆ QUICK RECALL QUESTIONS

1. What is Amazon S3?
 2. What is object storage?
 3. Why are storage classes important?
 4. What risks exist with public buckets?
 5. Why does governance matter in cloud storage?
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◆ FINAL ENGINEERING PRINCIPLE

“Modern cloud systems depend heavily on secure, scalable, and intelligently governed storage architecture.”